

YOLOv11-ASD: Improved YOLOv11 for Multi-Scale and Small Object Detection

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Abstract—Object detection plays a vital role in autonomous driving, where achieving both high precision and computational efficiency is a major challenge. To address the limitations of YOLOv11 in feature representation and computational cost, we propose YOLOv11-ASD, an enhanced architecture integrating three lightweight modules: ADown, SPPF-SENetV2, and DySample. ADown optimizes downsampling by reducing redundant convolutional operations; SPPF-SENetV2 enhances global receptive fields and channel-wise attention; DySample introduces a dynamic point-based upsampling mechanism at lower computational cost. Experiments on the KITTI dataset show that, compared to the original YOLOv11, YOLOv11-ASD improves mAP@0.5 by 2.4% (reaching 93.5%), reduces parameter size 19%, and lowers GFLOPs by 19.2%. The results demonstrate that the proposed model achieves higher detection accuracy and faster inference, making it highly effective for real-time object detection applications.

Keywords—yolov11; object detection; autonomous driving; yolov11-asd

I. INTRODUCTION

Autonomous driving is rapidly advancing as a key technology for the future of transportation. One of the core challenges in autonomous driving systems is enabling vehicles to reliably detect and interpret their surroundings in real-time^[1]. Object detection plays a crucial role by helping vehicles identify pedestrians, other vehicles, obstacles, and traffic signs to ensure safe navigation in dynamic environments. In recent years, deep learning-based algorithms like YOLO have become popular for real-time object detection due to their balance of speed and accuracy^[2]. YOLOv11, in particular, has been praised for its high inference speed, making it suitable for time-sensitive applications like autonomous driving^[3]. However, despite its advantages, YOLOv11 faces challenges, particularly in accurately detecting small objects and efficiently handling multi-scale features. Additionally, traditional downsampling and upsampling techniques can lead to computational inefficiencies, posing a challenge in real-time scenarios where speed and resource optimization are critical. To address these limitations, recent work has focused on improving detection accuracy and efficiency by optimizing key aspects of the detection pipeline. This includes refining the downsampling process, enhancing multi-scale feature extraction, and developing more efficient upsampling methods^[4]. These improvements are essential for enhancing the robustness of detection models in complex driving environments, where

variable lighting, occlusions, and rapid movement complicate the task^[5].

II. IMPROVEMENT OF YOLOv11 ALGORITHM

A. Introduction to the YOLOv11-ASD

Ultralytics' newly released YOLOv11, despite architectural enhancements and optimized training^[6], still faces challenges in achieving high precision and computational efficiency, especially in small object detection. YOLOv11 - ASD improves YOLOv11 by adding three modules: ADown for efficient downsampling, SPPF - SENetV2 for better attention - based feature aggregation, and DySample for dynamic upsampling. Its overall structure follows the common backbone - neck - head design, optimizing multi - scale feature handling and reducing computational load. The architecture is shown in Figure 1.

ADown replaces strided convolutions, reducing parameters while maintaining spatial information. SPPF-SENetV2 integrates channel-wise attention with multi-scale pooling for improved feature selection. DySample replaces conventional upsampling layers with dynamic interpolation, improving feature alignment. These changes result in a more efficient model with improved detection performance.

B. Lightweight ADown Module

Downsampling in standard YOLO architectures is typically handled by strided convolutions, which are fast but tend to discard valuable spatial information. ADown was introduced as a more efficient alternative, maintaining contextual information while reducing computational cost^[7].

The module leverages depthwise separable convolutions, followed by pointwise projections, with a residual shortcut to stabilize training and enable shallow feature reuse. Compared to traditional methods, ADown reduces the parameter count by over 20% without sacrificing representation power. It is applied consistently across all downsampling stages. The structure is shown in Figure 2.

C. SPPF-SENetV2 Attention Mechanism

To enhance the network's ability to integrate multi-scale features, the original SPPF module is extended with a channel attention mechanism, resulting in the SPPF-SENetV2 module. This improved block retains the fast multi-branch aggregation of SPPF while incorporating squeeze-and-excitation operations to recalibrate channel-wise feature responses. The design

effectively emphasizes informative features and suppresses irrelevant ones, especially in complex scenes with dense object distributions^[8]. The SPPF-SENetV2 module uses convolutional branches with varying receptive fields to capture spatial context

at multiple scales. A global average pooling operation is followed by a lightweight gating mechanism that assigns dynamic weights to each feature channel. Figure 3 illustrates the structures of the SENet, ResNeXt, and SENetV2 modules.

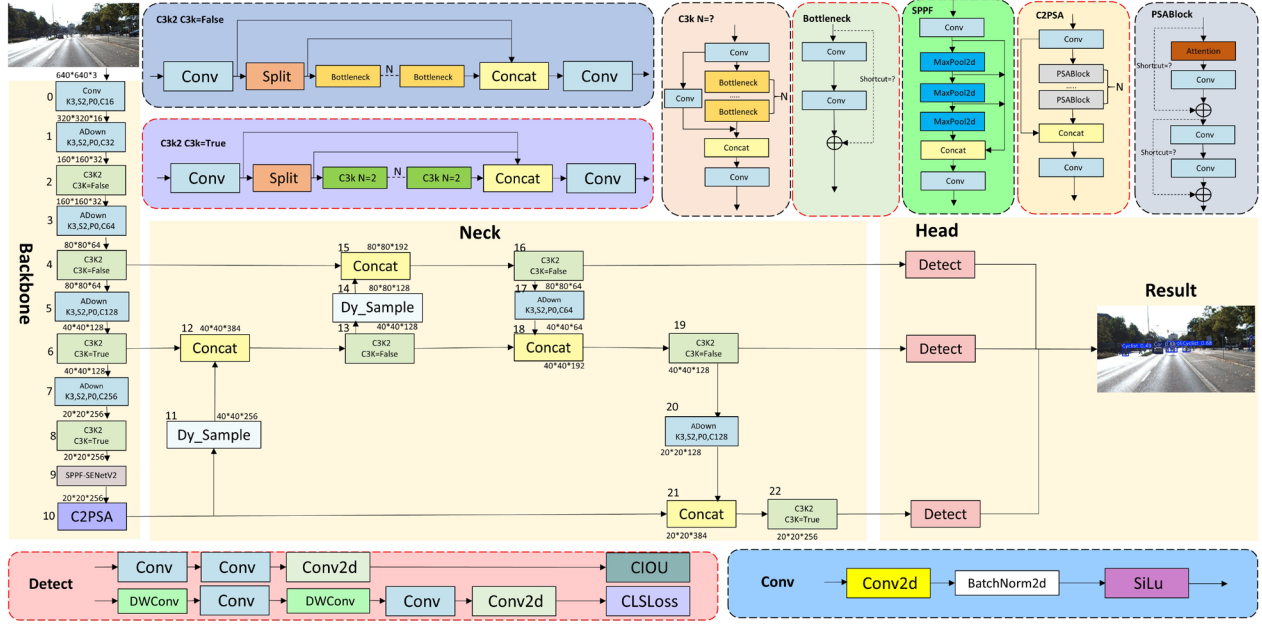


Figure 1. Network structure of YOLOv11-ASD

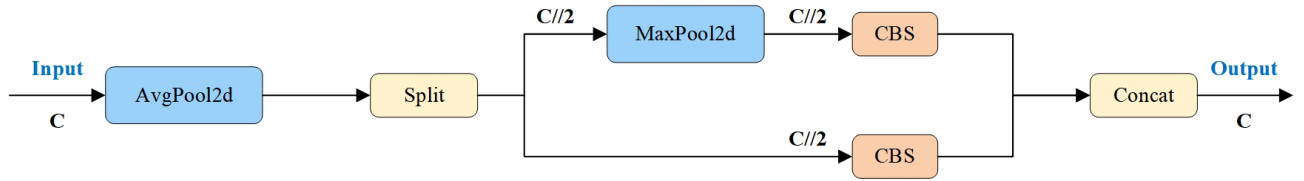


Figure 2. ADown Network Architecture

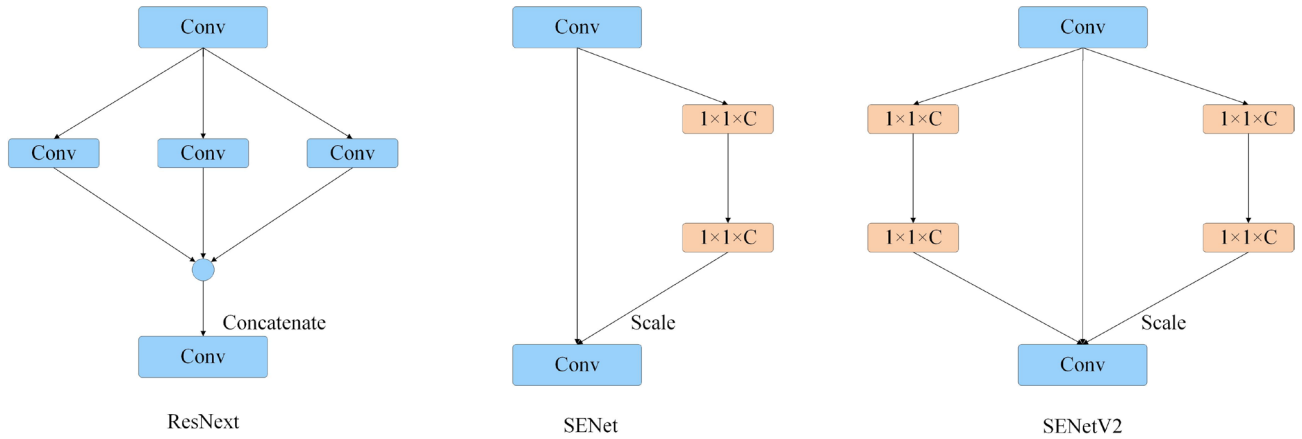


Figure 3. Comparison of ResNeXt, SENet, and SENetV2 Modules

D. DySample Module

Upsampling is often an overlooked stage in detection models, yet it plays a key role in aligning features at different resolutions. Most YOLO variants rely on nearest-neighbor or bilinear interpolation, which are computationally cheap but inflexible. These methods fail to adapt to variations in object scale or structure.

DySample addresses this by introducing a dynamic sampling strategy^[9]. Rather than using fixed interpolation kernels, it learns offset and weight maps from the input feature map and uses them to sample high-resolution features in a content-aware manner. Figure 4 shows the architecture of this module. Lightweight and fully differentiable, DySample improves the reconstruction of fine details—particularly for

small and irregularly shaped objects — without introducing significant computational overhead.

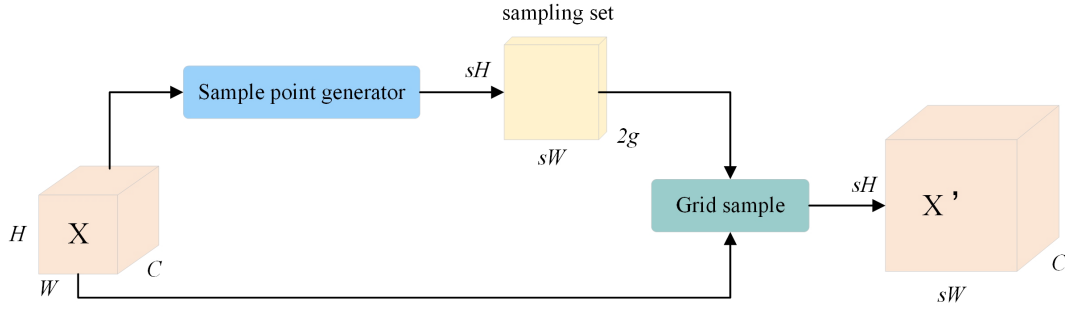


Figure 4. Architecture of the DySample Upsampling Module

III. EXPERIMENTAL ANALYSIS

A. Experiment setup

The experiments were conducted on a Windows 10 operating system, equipped with an Intel Core i7-12700KF

CPU and an NVIDIA GeForce RTX 2080Ti GPU (11GB VRAM), with CUDA 11.7 installed. The program was developed using the PyTorch 2.0.1 framework. The experimental parameters are summarized in Table I.

TABLE I. EXPERIMENTAL PARAMETERS

Experimental parameters	Parameter quantity
Epoch	200
Batch size	16
Learning rate	0.01
Momentum	0.937
Input size	640

B. Production of dataset

This study uses the widely adopted KITTI^[10] dataset, comprising 7481 images representing various real-world road and vehicle scenarios in autonomous driving research. The dataset was split into training, validation, and test sets at an 8:1:1 ratio. For annotation, five typical traffic participants were selected as detection targets: Cyclist, Truck, Van, Tram, and Car. To improve robustness against varying scales and occlusions, data augmentation techniques such as random cropping, brightness adjustments, and affine transformations were applied, enhancing the model's generalization ability.

C. Ablation Study

In this section, we conducted a series of ablation experiments on the pre-split KITTI dataset to further validate the contribution of different modules in YOLOv11-ASD to its performance. By progressively removing or replacing key components within the algorithm, we were able to assess the impact of each module on the overall detection accuracy, computational cost, and model parameter count. This approach allows us to gain a deeper understanding of the advantages of YOLOv11-ASD over the baseline YOLOv11 model, while demonstrating that the proposed improvements effectively enhance the performance of the algorithm.

TABLE II. ABLATION STUDY.

Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Flops (G)	Params (M)
YOLOv11	91.1	69.5	6.3	2.6
YOLOv11+ADown	92.3	70.7	5.1	2.1
YOLOv11+SPPF-SENetV2	93.2	72.2	6.3	2.6
YOLOv11+DySample	92.8	71.0	6.3	2.6
YOLOv11+ADown+SPPF-SENetV2	93.0	71.4	5.3	2.1
YOLOv11-ASD	93.5	72.6	5.1	2.1

As shown in Table II, the performance of the improved models in terms of mAP@0.5 and mAP@0.5:0.95 is higher than that of the baseline YOLOv11 model, with YOLOv11-ASD achieving the best results. Specifically, YOLOv11-ASD improves the mean Average Precision (mAP@0.5) by approximately 2.4% and mAP@0.5:0.95 by 3.1%. Furthermore, it reduces the false positive rate (FPR) and false negative rate (FNR) for small objects. The computational complexity (Flops) is reduced by approximately 19%, and the model parameters are decreased by 19.2%. These

improvements make YOLOv11-ASD more efficient for deployment in real-time systems.

D. Comparative Experiments and Experimental Results

To validate the superiority of the proposed improved algorithm over other methods, this paper compares the performance of YOLOv11-ECA with other mainstream YOLO series object detection algorithms using the KITTI dataset. The results of the comparative experiments are shown in Table III.

TABLE III. COMPARATIVE EXPERIMENTS.

Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Flops (G)	Params (M)
YOLOv8	92.4	71.1	8.2	3.0
YOLOv9	91.2	67.3	10.7	2.6
YOLOv10	89.4	68.6	8.3	2.7
YOLOv11	91.1	69.5	6.3	2.6
S-PANet-tiny	84.8	62.3	5.7	3.8
YOLOv11-ECA	91.8	71.4	6.3	2.6
YOLOv11-ASD	93.5	72.6	5.1	2.1

As shown in Table III, YOLOv11-ASD outperforms other mainstream algorithms in both mAP@0.5 and mAP@0.5:0.95. Specifically, it achieves an mAP@0.5 of 93.5% and an mAP@0.5:0.95 of 72.6%, surpassing all other models. YOLOv11-ASD also reduces computational complexity

(Flops) to 5.1G and model parameters to 2.1M. These improvements make YOLOv11-ASD more accurate and efficient, making it ideal for real-time applications where both accuracy and computational efficiency are critical.

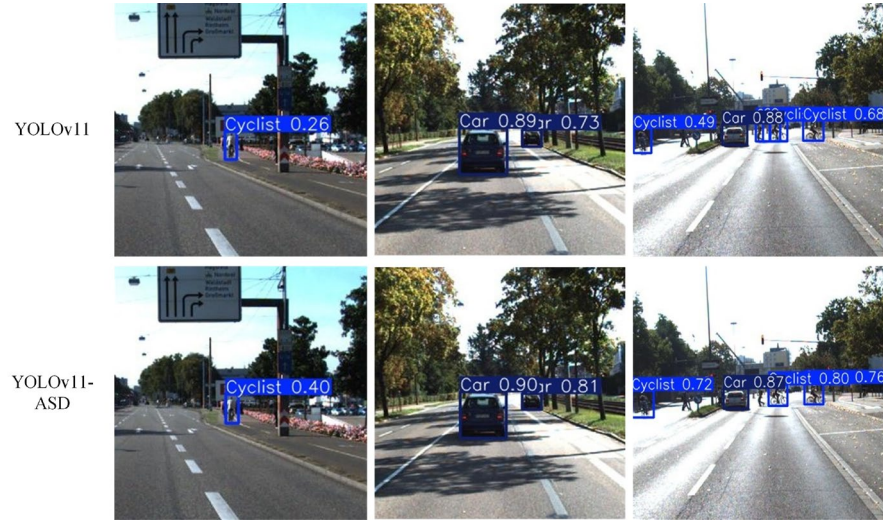


Figure 5. Comparison of test results

Figure 5 presents the detection results of the proposed algorithm in real-world scenarios, comparing them with the detection results of YOLOv11. By observing three images from different environments, it is evident that in each scene, YOLOv11-ASD consistently achieves higher confidence scores for the detected objects compared to YOLOv11. Notably, in a scenario where YOLOv11 misclassifies a person as a cyclist, YOLOv11-ASD correctly identifies the person, demonstrating its improved performance in distinguishing between similar objects. This indicates that YOLOv11-ASD outperforms the traditional YOLOv11 algorithm in terms of overall detection accuracy, particularly in complex environments. YOLOv11-ASD is better equipped to handle multi-scale objects in challenging backgrounds, showcasing its superior robustness and accuracy in real-time object detection tasks.

IV. CONCLUSION

To address the limitations of YOLOv11 in handling multi-scale object detection and feature representation under complex scenarios, this paper proposes an improved algorithm named YOLOv11-ASD. In the backbone stage, an ADown module is introduced to optimize the downsampling process and preserve more shallow feature information. At the end of

the backbone, the SPPF-SENetV2 structure is incorporated to enhance multi-scale feature fusion and integrate an attention mechanism for better focus on key regions. Additionally, multiple DySample modules are employed in the neck to replace conventional upsampling, improving feature recovery and significantly enhancing the detection of small objects. The proposed YOLOv11-ASD is trained and evaluated on the KITTI dataset. Experimental results show that it achieves a mAP@0.5 of 93.5%, representing a 2.4% improvement over YOLOv11. Furthermore, FLOPs and the number of parameters are reduced by 19.0% and 19.2%, respectively. These results demonstrate that YOLOv11-ASD improves detection accuracy while reducing model complexity, validating its effectiveness and potential for real-time object detection tasks such as autonomous driving.

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